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The role of generative AI in improving the sustainability and efficiency of HR recruitment process

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Abstract
Generative artificial intelligence (GAI) is becoming increasingly important in business processes, including human resource management. GAI can offer the potential to automate repetitive tasks in recruitment processes, optimise decision making, and reduce administrative burdens. Although AI can help increase operational efficiency, environmental pressures must also be taken into account. AI models require significant computing power, resulting in high energy consumption and increased CO_2 emissions. This dichotomy may raise the question of whether the efficiency gains provided by GAI outweigh the environmental burden. This article examines the environmental impacts of GAI on HRM through a case study. The research combines qualitative and quantitative methods: expert interviews are used to explore practical applications, while calculations on energy consumption, costs, and emissions are carried out by comparing traditional and AI-based recruitment methods. The results of the case study showed that the integration of GAI led to efficiency gains. The time required for the recruitment process was reduced by 13.25 h, which could save thousands of man-hours per year. At the same time, costs and energy consumption and associated carbon emissions were reduced. The study highlights the duality of "AI for sustainability" and "sustainability of AI", highlighting that while GAI can contribute to more sustainable corporate operations, its own environmental footprint raises questions about long-term sustainability. The results will provide HR professionals, decision makers, and organisations with practical insights into the potential for sustainable use of AI.

Keywords Generative artificial intelligence (GAI), Human resource management (HRM), Efficiency optimisation, ChatGPT, Sustainable development goals (SDGs), Interview, Westerm Transdanubia, Case study

1 Introduction

The market size of the artificial intelligence (AI) market in 2024 exceeded $184 billion, a jump of nearly $50 billion compared to 2023. However, this growth is expected to continue in the future and is expected to exceed $826 billion in 2030 [1]. The generative AI (GAI) market is also expected to grow significantly and is expected to grow by $320

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billion between 2024 and 2030, representing a + 887.41% increase. Thus, by 2030 the size of the GAI market could reach $356.05 billion [2].

GAI refers to a technology that can be used to create new content, including text, images, or sound. There are models that use natural language processing (NLP), such as the GPT Generative Pre-trained Transformer (GPT) models (e.g. GPT-3 and GPT-4) or BERT-based generative models. Examples of GAI models designed to develop particularly images are DALL-E, Stable Diffusion, or MidJourney. In addition to image generation, several models have been launched to generate sounds and videos, such as the Jukebox developed by OpenAI or the Runway Gen-2 for video generation [3]. Goldman Sachs (2023) reports that GAI can increase global gross domestic product (GDP) by up to 7%, which could replace up to 300 million jobs of knowledge workers [4].

The rapid uptake of GAI is also due to the fact that many industries have started to use it and integrate it into their processes. The healthcare sector has had the most significant impact on medical research, diagnosis, treatment, and patient care, and has already made significant progress in real-world applications [5]. Given the potential of technology to assist medical professionals in decision making, Introzzi et al. investigate how GAI tools perform in clinical practice and whether they have achieved optimal integration with human decision makers. To investigate this, they analysed a specific scenario of AI assistance during colonoscopy in their investigation [6]. The financial sector also sees great potential in technology, as it has the ability to generate financial forecasts, thereby also aiding decision making. In their case study, Li et al. (2024) investigated how GAI can provide financial forecasts to financial professionals and concluded that ChatGPT was able to reduce the optimism bias observed in financial forecasts of financial forecasts [7]. In addition to these industries, marketing [8], business management [9], education [10], or data science [11], among others, could be other relevant fields.

AI and GAI have great potential in all fields and industries, but this technology also has its own environmental concerns. It is questionable whether, with its rapid growth, AI will have a positive impact to offset its environmental impact and harmful pollution. In 2023, AI's energy consumption worldwide is estimated at 4.5 gigawatts, which is 8% of the energy consumed by data centres in that year. However, it is predicted that by 2028, the amount of energy consumed will triple, approaching 20%. However, this share would not be evenly distributed worldwide, as, for example, while in the US, data centre energy consumption could account for 6% of the country's total energy consumption by the middle of the decade, in Ireland it could be up to a third [12].

AI is playing an increasing role in the sustainability goals. Several of the Sustainable Development Goals (SDGs) of the UN are closely related to AI applications, such as decent work and economic growth (SDG 8), the development of industry, innovation, and the infrastructure (SDG 9), and action on climate change (SDG 13) [13]. Technology offers opportunities to increase efficiency, optimise energy use, and support environmentally friendly solutions. At AI same time, the operation of AI has a significant environmental burden, which is measured by the high energy demand of data centres, the computational capacity required to train models, and the associated CO_2 emissions [14].

In the current literature, there are many studies on the sustainability benefits of AI, but few studies have examined the positive and negative impacts in parallel. The novelty of this paper lies in the fact that it not only analyses the sustainability benefits of AI, but also discusses the ecological footprint of the technology in a comprehensive way. As

AI becomes increasingly embedded in industry processes, it is essential to take a holistic approach to assessing sustainability and understanding the extent to which AI supports or hinders the achievement of sustainability goals. This research aims to balance this dual perspective, taking into account the interlinkages of AI applications with the SDGs. On the one hand, we explore how GAI can help optimise business processes and promote sustainability. On the other hand, we assess the environmental costs associated with the use of AI models, with a particular focus on energy use and greenhouse gas emissions.

The main objective of the present research is to investigate the sustainability and environmental impact of integrating GAI into different corporate processes, in this case in the area of recruitment. To investigate this, calculations have been carried out using a case study, looking at efficiency in terms of costs and working time, as well as emissions in terms of carbon dioxide equivalent ($CO_{2\text{-eq}}$). In this paper, we investigate the application of GAI in recruitment processes through a case study, assessing the efficiency and cost-reducing potential of the technology. Although the concept of sustainability includes environmental, social, and economic dimensions, this study focusses specifically on environmental impacts, particularly energy consumption and CO_2-equivalent emissions, related to the integration of generative AI in recruitment.

In line with the aim of the article, two research questions have been formulated, which we aim to answer through literature, interviews with experts, and, most importantly, the case study.

- (RQ1) In which tasks in the HR recruitment process can GAI be applied to increase efficiency and reduce costs?
- (RQ2) How does the use of generative AI in recruitment affect the environmental performance of HR processes in terms of energy consumption and emissions?

Although a significant number of recent studies explore the potential of GAI to influence sustainability outcomes, the majority of these works remain conceptual in nature, focusing on theoretical models or high-level industry overviews. There is a notable absence of empirical and quantitative analyses that assess the environmental impacts of GAI technologies within concrete organizational settings and operational business processes, such as energy usage, $CO_{2\text{-eq}}$ emissions, and cost efficiency. In particular, studies that simultaneously examine both AI for environmental performance improvement and the environmental footprint of AI systems themselves within a single empirical case remain scarce. This gap in the literature underlines the relevance and contribution of the present study.

The novelty of this research lies, on the one hand, in examining the quantitative environmental impacts (eg $CO_{2\text{-eq}}$ emissions and costs) of the concrete enterprise-level application of GAI. Furthermore, our case study projects the effects of GAI use on real HR processes, using business process modelling (BPM) and computational emissions and cost analysis. This complex, multidisciplinary approach makes the study unique, providing not only theoretical but also practical guidance at the intersection of HR and sustainability.

The novelty of this research lies in its quantitative assessment of the environmental and economic impacts, such as $CO_{2\text{-eq}}$ emissions and operational costs, resulting from the enterprise-level application of GAI. The case study examines real-world HR

recruitment processes, applying business process modeling (BPM) alongside data-driven analysis of energy use and expenditures. This integrative and multidisciplinary approach enhances the originality of the study, offering both theoretical insights and actionable guidance at the intersection of human resource management and environmentally conscious digital transformation.

2 Theoretical background

The Theoretical Background section primarily provides a general overview of generative AI and related ethical considerations. This is followed by a presentation of the potential applications of GAI in HRM and the results of related empirical research. The rest of the section examines sustainability aspects from two perspectives: AI for sustainability and the sustainability of AI. Table 1 summarizes the studies that laid the foundation for the present research. Based on the literature, we have concluded that the research on AI lacks a two-sided approach, and that it is necessary to examine how AI can contribute to environmental sustainability when applied in organizations, but also what environmental impact it has. The studies and other research in the table are discussed in more detail later in this section.

2.1 Generative artificial intelligence (GAI) and ethical concerns

One of the fastest-growing areas of computer science is AI, which is concerned with the development of systems and algorithms that mimic human intelligence. AI aims to implement automated decision making, problem solving and pattern recognition in various application domains [27]. The structure of AI can be divided and structured into several main branches, among which machine learning (ML), natural language processing (NLP), computer vision, robotics, speech recognition, and GAI stand out [28].

ML is a key branch of AI that allows computers to learn from data and improve their performance autonomously, without being specifically programmed to do so. ML

Table 1 Overview of reviewed studies related to GAI, HRM, and AI sustainability

Topic	Authors	Main results
Generative AI in human resource management	Chuang et al. [16]	Data-driven model enhances objectivity and transparency in AI-supported recruitment decisions
	Pan et al. [17]	Technological complexity hinders AI adoption; regulatory support encourages it
	Abdelhay [18]	GAI improves recruitment accuracy and bias reduction; position level mediates effectiveness
	Hewage [19]	AI suitable for sourcing, pre-screening, and candidate engagement; limited use in interviews
Sustainability of AI	Patterson et al. [20]	Training large AI models emits significant CO_2, requires mitigation strategies
	Li et al. [21]	GPT-3 inference may consume 500 ml water per 10–50 responses
	Henderson et al. [22]	Proposed standardised framework to measure ML training emissions; promotes transparency and energy-efficient practices
	Ligozat et al. [23]	LCA reveals overlooked lifecycle emissions in AI; most studies focus only on usage phase
AI for sustainability	Nishant et al. [24]	AI promotes sustainability but requires balanced governance to mitigate ethical and environmental risks
	van Wynsberghe [25]	Lack of lifecycle perspective in AI sustainability; proposes Sustainable AI as new framework
	Khakurel et al. [26]	AI affects all five sustainability dimensions both positively and negatively

Sources: own edit

algorithms can be classified into three main categories: (1) supervised learning, (2) unsupervised learning and (3) reinforcement learning. These algorithms are widely used for applications such as making predictions, classification tasks, and running recommender systems [29]. NLP is an AI technology that enables the interaction between computers and human languages. Applications of NLP include machine translation, chatbots, automatic text parsing, and intelligent search systems [30]. Computer Vision allows computers to interpret visual information by analysing images and videos. Applications of computer vision include face recognition, autonomous vehicle navigation, and medical imaging, where AI can help diagnose diseases [31]. Robotics, by combining AI and engineering, aims to create autonomous systems that can sense their environment and make appropriate decisions. Intelligent robots are also playing an increasing role in industrial automation, healthcare, and logistics [32].

GAI is one of the newest and most dynamically evolving branches of AI, capable of generating new and original content, such as text, images, music, or even video. Models such as GPT (Generative Pre-trained Transformer) and DALL-E have revolutionised the creative industries, enabling automated content creation [3]. AI has created opportunities and challenges in a variety of fields, including business, education, healthcare, and the arts and humanities [33]. GAI can serve businesses and organisations in a variety of ways, ranging from marketing, engineering, risk and legal, human resources, accounting, and financial forecasting [34].

Ethical concerns refer to the moral obligations and duties of an AI application and its creators [35]. The use of AI in recruitment processes raises a number of ethical issues, including transparency, data protection, and the marginalisation of the human factor. According to a study by Sýkorová et al. [36], the opacity of algorithmic decisions ("black-box effect") in AI-based recruitment can pose a significant risk, making it difficult to explain selection decisions and therefore undermining candidates' trust. The authors also argue that it is particularly critical that candidate data is often processed without their consent, which is contrary to the provisions of the GDPR and the new AI Act [36]. Lukács and Váradi [37] emphasise that automated decisions, especially when not backed by meaningful human intervention, can have a significant impact on worker rights and status and may even have discriminatory consequences. Although the applicability of Article 22 of the GDPR is limited, even in cases of partial automation, the principles of lawfulness, necessity, and human review must be complied with. Algorithmic decision-making poses particular challenges in the field of HR, as such decisions often have a fundamental impact on employees' career paths and can also have significant consequences at the societal level. AI-based systems often lack transparency, that is, it is not clear what factors the system uses to make decisions, which can undermine employee trust and make legal accountability difficult [38].

Harmful content created by GAI may include material that is violent, discriminatory, offensive, or pornographic. Although OpenAI applies a content policy to ChatGPT, such content may still be displayed due to algorithmic restrictions or jailbreaking (unblocking restrictions). The ability of language models to generate offensive or harmful content is referred to as toxicity, which can compromise social harmony. Therefore, it is of paramount importance to clean training data and remove harmful information [34]. The concept of bias refers to the possibility that AI-generated responses or recommendations may unfairly favour a person or group [39]. Bias can sometimes be observed in the

content generated by language models, which may result from training data. For example, exclusionary norms are created when data are representative of only a narrow group of society. Similarly, multilingualism bias occurs when the model is trained exclusively in a single language. Given that ChatGPT is globally available, taking into account cultural sensitivity is the key to minimising biases [40]. Hallucination is perhaps one of the most talked about and widely known limitations, which means that the model generates output that is meaningless, untrue, and inappropriate for the source input. This is a widely recognised limitation of GAI and can include textual, auditory, visual, or other types of hallucination [41]. GAI can generate seemingly correct answers but without any meaning [34].

2.2 AI applications in human resource management

AI tools and digital platforms have become an indispensable part of organisations and society as a whole over the past decade and the impact can be observed in the field of human resource management [42]. The application of AI in HRM is seen in two main ways: in the field of augmentation (human decision support) and in the field of automation (human job augmentation) [43]. Recent research has highlighted that AI-based decision support systems (DSS) often lack the level of trust and transparency needed for confident use in business contexts. For example, Hamrouni et al. [15] developed a taxonomy and prototype of AI-supported business model tools that aim to increase explainability and user trust, an issue equally relevant when integrating GAI into HR decision making processes.

AI can help increase the efficiency of various HRM tasks through different ML and deep learning (DL) models and during the employee lifecycle. The employee lifecycle consists of five main phases (1) Recruitment, (2) On-Boarding, (3) Employability and Benefits, (4) Retention, and (5) Off-Boarding [44]. In the area of recruitment, AI can help in selecting the right employee by using machine learning-based algorithms (e.g. Decision Tree, Random Forest, KNN) [16]. In the On-Boarding phase, AI can be used to offer personalised training programmes to facilitate employee development (eg, neural networks, adaptive boosting) [45]. In the next stage of life, the Random Forest (RF) and Artificial Neural Networks (ANN) models, among others, can help predict employee progression and performance [46]. In the fourth stage of life, the Retention stage, the focus is on optimising the work environment, predicting stress and mental workload, for which support vector machine (SVM) models can be effectively applied [47]. In the last stage, the Off-boarding stage, AI models (eg XGBoost, Logistic Regression) can help predict employee turnover and identify its causes [48].

Pan et al. examine the application of AI in recruitment, based on the technology, organisation and environment (TOE) model and transaction cost theory. AI-based tools in recruitment can help identify and evaluate potential employees. AI chatbots and virtual assistants can help HR teams connect with candidates [17]. Furthermore, AI enables data-driven decision making, reducing subjective bias in selection. Research among Chinese companies found that the adoption of AI is most hindered by technological complexity, while technological competence and regulatory support encourage it. The size and industry sector were not found to be determinants, but the transaction costs influence adoption. User-friendly technology, skilled workforce, and a regulatory environment that respects AI are important for successful AI adoption [17]. Abdelhay

[18] empirically examined the use of ChatGPT in optimising the recruitment process, involving 469 professionals. The research used structural equation modelling (SEM) to confirm that GAI tools contribute to reducing bias, increasing efficiency, and screening candidates more accurately. User knowledge of AI played a moderating role in perceived effectiveness, while organisational hierarchy (e.g., position level) influenced the results as a mediating factor in some cases. This study significantly reinforces the findings of Pan et al. (2021), while also introducing new factors, such as organisational context, into the interpretation of the use of GAI in recruitment. Abdelhay et al. [49] found that process automation plays a mediating role in efficiency, while organisational size only moderately influences the effect of GAI on recruitment. According to their findings, higher levels of automation significantly enhance the positive effects of generative AI, especially in larger companies, but the agility of smaller organisations can also be an advantage in introducing the technology. Hewage's [19] qualitative research is based on in-depth interviews with 17 HR experts and examines the applicability of artificial intelligence in various recruitment phases, particularly in the recruitment pre-planning, sourcing, pre-screening, interview, and candidate engagement stages. According to the results, AI is most successfully used in the pre-screening and sourcing phases. When it comes to interviews, the vast majority of professionals prefer human evaluation, primarily due to the lack of empathy and situational judgment. This approach is consistent with the empirical findings of Abdelhay (2024) [18] and Abdelhay et al. [49] that GAI tools can be used effectively in the early stages of pre-screening and selection, but that the human factor remains essential in final decisions. Horodyski [50] conducted a study on a sample of 283 HR professionals from 15 countries to examine the factors that influence the acceptance of AI tools in recruitment processes. The analysis was based on the UTAUT model and showed that performance expectations have a positive, statistically significant effect on the intention to use artificial intelligence. Effort expectations and social influence were not found to be significant. The results are consistent with the studies by Abdelhay [18] and Hewage [19], which also emphasised the role of efficiency, time savings, and automation in the adoption of generative AI for recruitment purposes. Khan et al. [51] highlight that generative AI significantly improves HR efficiency, particularly in recruitment and performance management, while also raising important ethical issues such as algorithmic bias, privacy, and lack of emotional intelligence. Their literature-based study recommends proactive governance frameworks to ensure fair and responsible implementation of GAI in HR practices.

ChatGPT and similar AI technologies bring many benefits to HRM. One of these areas is recruitment and selection, where ChatGPT is able to quickly review large volumes of CVs and select the most suitable candidates, thus reducing recruitment time and costs [52]. With the help of AI, position-specific interview questions can be created to facilitate a more thorough assessment of candidates. Other areas of HRM could include performance appraisal, employee training, administration, and data management. ChatGPT-based chatbots provide 24/7 support to employees, answering HR-related questions, and helping to solve problems. ChatGPT can help create and update HR documents such as employment contracts and policies, ensuring that they are accurate and up to date [52]. According to a study by Aguinis et al. (2024), GAI has the potential to alleviate HR professionals' overload and help them with both operational and strategic tasks. The use of GAI in eight key HRM areas—such as recruitment, performance management, benefits

management, employee retention, and strategic HRM—increases efficiency, supports decision-making, and helps improve the employee experience. AI can automatically process CVs, generate recruitment advertisements, offer personalised training programmes, and predict turnover risks. The article stresses that the key to successful AI deployment is the use of appropriate prompts and careful monitoring of the responses generated, as AI cannot replace human expertise on its own, but can complement it [53].

2.3 Sustainable AI

Although AI can contribute to several SDGs and support the achievement of sustainability goals, for example by providing different predictions, the technology itself can also be polluting due to its high energy consumption [54]. However, in general, it is impossible to say whether AI tips the scales in the positive or negative direction.

2.3.1 Sustainable development goals

The Sustainable Development Goals (SDGs) are global targets set by the United Nations to achieve a more just, sustainable, and environmentally friendly world. The SDGs comprise 17 main goals and 169 specific targets, building on previous MDGs but expanding on them to address new challenges [55]. These goals include eradicating poverty, reducing inequalities, combating climate change, access to clean water and sanitation, and equitable and sustainable economic development. The SDGs provide a comprehensive policy framework that allows countries to align their own development strategies with them, while ensuring the selection and use of relevant indicators with scientific and expert support [55].

AI is playing an increasing role in achieving sustainability and supporting the Sustainable Development Goals (SDGs) of the UN. The use of AI can help optimise the use of natural resources and energy, while improving environmental management processes and reducing the ecological footprint of human activities [24]. AI can use big data analytics and machine learning algorithms to predict sustainability challenges, helping decision makers develop more effective environmental strategies. However, the application of AI also poses challenges, including privacy concerns, cybersecurity risks, and ethical issues that require an appropriate regulatory framework. To ensure long-term sustainability, it is essential that the development of AI follows a multidisciplinary approach, taking into account not only the technical potential, but also social and economic factors [24].

AI can make a significant contribution to achieving the SDGs. For example, the application of AI in healthcare can improve diagnostic accuracy, contributing to the achievement of the goal of "Good Health and Well-being" (SDG 3). In education, AI can offer personalised learning experiences, supporting the goal of "Quality Education" (SDG 4). By optimising industrial processes and developing smart infrastructures, AI can contribute to the goal of "Industry, Innovation, and Infrastructure" (SDG 9). Furthermore, AI can help anticipate the impacts of climate change, supporting the objective of "Action against Climate Change" (SDG 13) [56].

Many of the SDGs are directly or indirectly related to HRM, but the chances of achieving the goals are reduced by the lack of clarity on the concept of HRM and the challenges inherent in the nature of the goals. According to Brewster & Brookes (2024), HRM cannot promote the achievement of the goals due to a lack of conceptual clarity. However,

the authors stress that the SDGs can be valuable for HRM practitioners as they draw attention to the importance of multiple stakeholders and the key role of HRM in organisations [57].

Sustainable Human Resources Management (SHRM) is a key element in achieving the Sustainable Development Goals (SDGs) of the UN. According to Chams & García-Blandón, SHRM does not only focus on workplace sustainability, but can help organisations achieve the SDGs by introducing ecological techniques and integrating innovative sustainability strategies. SHRM is presented as a tool that can contribute to creating a healthy work environment and protecting individuals, communities, and the planet [58]. Companies must integrate environmental responsibility into their HR processes to achieve long-term sustainable growth, reduce their ecological footprint, and meet regulatory requirements [59].

2.3.2 Sustainability of AI and AI for sustainability

The development of AI has a revolutionary impact in many industries, but this technological leap comes at a significant environmental cost. GPT-3's water consumption during the training phase was estimated at approximately 4.8 billion litres of water, assuming the model was trained at Microsoft's data centre in Iowa. If the model had been trained entirely in the Washington data centre, water consumption could have been as much as 15 billion litres, which would have been more than Microsoft's total water consumption in 2023 [60]. AI models, especially LLMs, require a huge computing capacity, which involves not only significant energy consumption, but also a water footprint [21]. Most data centre cooling systems use water-based solutions, which consume huge amounts of water to train and run the models. In this context, a key issue for sustainable AI development is how to reduce the water requirements of AI systems without negatively impacting their performance. The development of green data centres, water-efficient cooling solutions, and the integration of renewable energy sources are all steps that can help mitigate the environmental impact of AI. Current trends show that technological advances and environmental concerns must go hand in hand [21].

AI models, in particular LLMs, require extreme computing power, which results in significant energy consumption. Models such as GPT-3, GPT-4, or Llama 2 consume huge amounts of electricity during their operation, directly contributing to carbon emissions and exacerbating climate change [20]. Based on research from Stanford University published in 2024, GPT-3 (175B) model (2020) has the largest $Co_{2\text{-eq}}$ emissions with 502 metric tons. Models with higher parameter counts tend to emit more significant CO_2, which is related to energy-intensive training processes and computational capacity [61]. It is estimated that AI will account for 4.5 gigawatts of energy consumption worldwide in 2023, which will represent 8% of total data centre energy consumption in that year. AI power consumption is expected to grow significantly over the next five years, with a projected 14–18.7 gigawatts in 2028. This could represent up to 20% of data centre power consumption in that year [62].

Many technology companies and research institutes are working on energy-efficient models to develop more sustainable AI. Some strategies include using optimised hardware, developing more efficient algorithms, and switching to renewable energy sources to power data centres [22]. While AI offers significant economic and social benefits, reducing energy consumption will be the key to long-term sustainability. Researchers

are currently working to develop low-carbon AI models that have a smaller ecological footprint while maintaining high performance [22].

Sustainable AI is not a precisely defined term and, therefore, means different things to different researchers and in different ways. However, Van Wynsberghe (2021) has proposed a definition of „sustainable AI", as follows: „Sustainable AI is a field of research that applies to the technology of AI (the hardware powering AI the methods to train AI, and the actual processing of data by AI) and the application of AI while addressing issues of AI sustainability and/or sustainable development. Sustainable AI does not exclusively with the implementation or use of AI but should address the entire life cycle of AI, the sustainability of design, training, development, validation, re-tuning, implementation, and use of AI." [25].

The literature on the sustainability of AI can be divided into two parts, there are studies that look at AI as a means of achieving sustainable goals and there are studies that analyse the impact of AI on sustainability (Fig. 1). However, studies often only look at one side of the spectrum, which does not give an accurate picture of what the sustainability of AI actually means [63]. Dhiman et al. (2024) [63] conducted a literature analysis and found that of the 88 articles examined, 43.2% dealt with the sustainability of AI, the same percentage dealt with AI for Sustainability and only 13.6% took a holistic approach, where both sides of the issue were examined [63].

Sustainability can be analysed by focussing on one of the economic, social, or environmental dimensions [64]. Khakurel et al. [26] identified five dimensions of AI sustainability. The economic dimension highlights that AI-driven automation can reduce the need for outsourcing, which can be a competitive advantage for developed economies, but also a threat to the labour markets and economies of developing countries. The technical dimension shows that the rapid development of AI is increasing the speed of R&D, but also raises ethical questions as developers' responsibility for the operation of AI systems increases. The environmental dimension includes both positive and negative impacts. AI can help protect the environment, for example in waste management or climate change prediction, but the increased technological consumption generated by AI can also lead to electronic waste and depletion of natural resources. The individual dimension shows that AI can reduce working hours and increase people's efficiency, but it can also cause anxiety and isolation due to increased human–machine interaction. Finally, the social dimension explores the impact of AI on communities, which includes the potential to strengthen social cohesion, but also the need for legal and ethical regulation, especially in relation to AI's responsibility and decision-making autonomy [26].

Life Cycle Analysis (LCA) is a widely recognised methodology for environmental impact assessment, with ISO standards (ISO 14040 and 14,044) and an ETSI/ITU specific methodological standard for ICT. Quantifies a number of environmental criteria and covers the different stages of the life cycle of the target system [65]. Quantifying

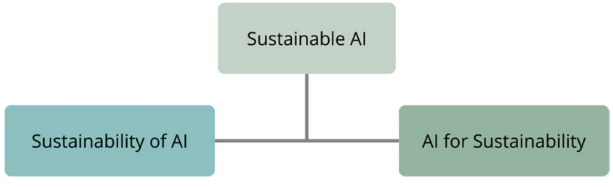

Fig. 1 Sustainable AI, own edit [25]

the impact of digital technologies requires a number of methodological decisions to be made, which define the exact framework of the system under evaluation. Assessing the global impact of AI, even if it could be accurately defined, is not the same as measuring the environmental impact of a specific AI algorithm or service. Quantifying the impact of AI is still an emerging research area, lacking a consistent methodology and focusing on the use phase of the tools involved in AI services [23].

Figure 2 shows the life cycle of the five phases of the AI system, each of which involves significant resource use and emission of resources. The first phase (01) is the extraction of raw materials, during which the rare earth elements, silicon, lithium, and cobalt, needed to produce AI hardware, are mined. This process not only involves large amounts of water consumption and pollution, but also has serious ecological impacts. This is followed by the (02) manufacturing and assembly phase, during which hardware components such as servers, cooling systems, and network infrastructures are produced. This is a highly energy-intensive process that releases toxic chemicals and significant amounts of greenhouse gases into the environment. Transport and distribution of the manufactured components and devices (03) create an additional environmental burden, as global supply chains generate significant fuel consumption and carbon emissions. The biggest ecological impact is in the (04) use phase, where the operation of AI models is a huge energy demand. The training of models, especially for deep learning algorithms, requires an extraordinary amount of computational power and, thus, electrical energy. In addition, data processing and storage, as well as inference (the day-to-day use of models), also result in continuous energy consumption, as data centres require constant operation. Finally, in the end-of-life phase (05), recycling or scrapping of obsolete AI hardware poses additional challenges. The proper processing of electronic waste and the recycling of raw materials is essential for more sustainable technological development, but currently these processes still have a significant environmental burden [23].

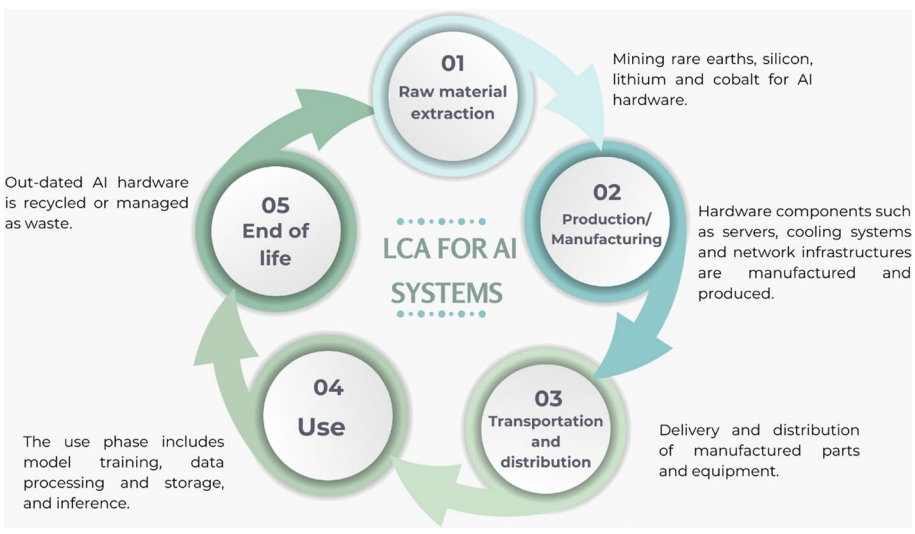

Fig. 2 Life Cycle of AI systems, own edit [23]

3 Methodology—case study

A case study was conducted to investigate whether the integration of GAI into a given business process can increase efficiency, its cost implications, and its environmental impact. In this case study, we used a recruitment process in an organisation to examine these aspects. Section 3 describes the in-depth interview with a HR professional, the BPM graph that was created using the qualitative data, and the methods and formulae used for the calculations. Figure 3 shows the methodology used in the case study, which will be explained in more detail in Sect. 3, Methodology—case study.

3.1 In-depth interview

3.1.1 Research approach

The present research can be interpreted as a single descriptive case study, as it examines the recruitment process of a single organisation in Hungary, with a particular focus on the integration of GAI. The purpose of the case study is not to make comparisons, but to provide a detailed description of how time, cost, and environmental indicators change in a specific organisational context following the introduction of ChatGPT. The case study focusses on an organisation that has officially integrated generative AI into its recruitment processes starting in 2024, providing an excellent example for exploring the practical application of this new technology. The descriptive approach allowed for the modelling and quantitative analysis of each step of the recruitment process.

3.1.2 Data collection

Data collection, the interview, lasted approximately 2 h, during which the interviewee identified 31 process steps in the recruitment process. In addition to process steps, an average time was associated with each step so that later calculations could be made for cost, energy, and $CO_{2\text{-eq.}}$ The interviewee also estimated how much time each task would take without technology and how much time with technology. All methods were

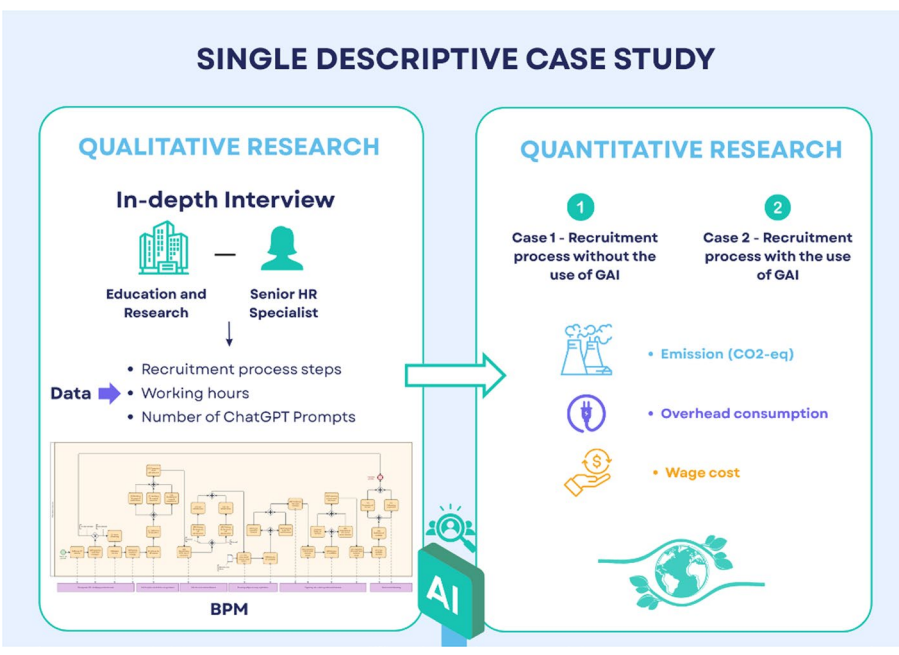

Fig. 3 Methodology, own edit

carried out in accordance with relevant guidelines and regulations, including the Scientific Ethics Code of Széchenyi István University. The study involved a single, anonymised expert interview focusing on general HR processes and AI usage, without the collection or storage of any personal or sensitive data. Prior to participation, the interviewee was informed about the purpose of the study and gave their explicit consent to participate voluntarily. No identifiable information was recorded, and the interview content has been fully anonymised. Table 2 shows the main characteristics of the interviewee.

The interview had three main objectives. First, we examine how the recruitment process is structured within the organisation, and, based on this, we prepared a BPM diagram showing the recruitment process. Second, we asked the interviewee to estimate the time required for each task according to two different scenarios: (1) using the traditional method without ChatGPT and (2) using ChatGPT. Third, we asked the interviewee to estimate the number of prompts used when ChatGPT was used. The interview was conducted according to a pre-prepared set of questions and was recorded. The interviewees were selected because they have expert knowledge of HR operations and direct experience with the organisational application of generative AI tools. After collecting the data, we provided the completed BPM diagram and time estimates to the interviewee, who provided feedback and approved the data, ensuring the accuracy and reliability of the information.

The outline of the interview was reviewed by experts. An HR professional with experience in AI applications and a researcher experienced in qualitative research methodology evaluated the relevance and completeness of the questions, ensuring that they covered the scope appropriate for the purpose of the research.

Interview Questions

Section 1—Mapping the Recruitment Process

- Can you describe the complete recruitment process in your organisation, step by step?
- Who is typically involved at each stage of the recruitment process?
- How long does each step usually take on average?
- Are there any formalised workflows or planning tools you follow (eg, annual HR planning)?

Section 2—Generative AI Usage

- Since when have you used generative AI tools (such as ChatGPT) in the recruitment process?
- At what specific steps do you integrate ChatGPT or similar tools?
- What tasks does ChatGPT support you with most effectively:
- Are there steps where you explicitly avoid using AI? Why?

Table 2 Interviewee profile

	Attribute	Description
1	Gender	Female
2	Position	Senior HR Manager
3	Organization Type	Higher education institution
4	Years of experienxe	6 years in HR
5	Sector	Education and Research
6	GAI usage	Regular use of ChatGPT since early 2024

Sources: In-depth interview, own edit

Section 3—Time Estimation

- How much time would each task take without using ChatGPT?
- How much time does each task take now with ChatGPT support?
- Can you estimate the average number of ChatGPT prompts that you use per task?

Section 4—Perceptions and Experiences

- What are the main benefits and challenges of integrating GAI into recruitment?
- Do you see further steps in the recruitment process where AI could be applied in the future?

3.1.3 Data analysis

The interview text was processed using thematic analysis, which allowed us to identify the main themes and patterns related to the steps of the recruitment process and the impact of AI use. Based on the complete transcript, the results were first processed using open coding and then analysed according to the three main objectives of the interview. These thematic categories included (1) the steps of the recruitment process, (2) the time required for each step of the recruitment process with and without ChatGPT, and (3) the estimated number of prompts for each step of the process. To model the recruitment process and develop the two types of time estimates, we performed a structured tabular analysis based on the steps identified in the interviews, which also allowed us to extract comparable quantitative inputs. The reliability of the thematic framework was validated by a second researcher by independent coding and comparison of the results.

3.2 Business process modelling—recruitment process

BPM was used to model the HR recruitment process studied in this research. This model was found to be the most suitable for visualising the process, as it allows the representation of the activities of a process, illustrating its logical sequence, and thus contributing to the achievement of corporate objectives [66]. The construction of the model was preceded by an in-depth interview, during which the BPM diagram was created using qualitative data and presented to the HR professional, who verified it with his professional knowledge. Although the aim of this research is not primarily innovation in HR process, but environmental aspects, it can provide useful information for HR professionals to make the recruitment process more efficient.

3.3 Environmental and cost calculation

3.3.1 Wage and overhead cost

Three types of costs have been identified, which can represent expenses and costs for the organisation. Wages and overheads have been calculated per hour worked to show the differences between the two scenarios. Overheads include a total of four elements (electricity, gas energy, District heating and hot water service charges, water and sewerage charges), the cost of which is calculated per person and per hour worked. The third type of cost is the ChatGPT subscription fee. In the present case study, the general ledger of an organisation was used to obtain the costs for the year 2022 for these four categories, as well as the number of employees in the organisation. These two data were used to determine the cost per hour for the four categories studied. The values were calculated in HUF and then converted into EUR at the exchange rate of 0.0025 EUR/HUF

on 26.02.2025 [67]. Since these are sensitive data for the organisation, such as overhead costs and number of employees, we do not publish them in this article, but the exact methodology is presented so that it can be applied to other companies.

(1) Overhead cost per person per hour

$$C_{o,p/h} = \left(\frac{C_{o,y}}{12 \times 160}\right) \div N_e$$

To determine the salary of an HR professional, an average hourly wage was calculated based on the Hays Hungary 2025 report. To calculate the wage cost, the average gross monthly wage was divided by 160 h, which corresponds to an average hourly cost [68]. The exact determination of wage costs is further elaborated in Sect. 5.

(2) Wage cost per hour

$$C_{w,h} = \frac{C_{w,m}}{160}$$

The ChatGPT subscription fee is 20 USD per month (02.03.2025), which, if multiplied by the number of subscriptions and 12 months, gives the cost of the service to an organisation for 1 year.

(3) ChatGPT subscription cost for 1 year

$$C_{GPT} = C_{GPT,m} \times C_{GPT,n} \times 12$$

where:

- $C_{o,p/h}$: Overhead cost per person per hour
- $C_{o,y}$: Annual cost of overhead
- N_e: Number of employees
- C_{GPT}: ChatGPT subscription cost for 1 year
- $C_{GPT,m}$: Monthly cost of ChatGPT subscription
- $C_{GPT,n}$: Number of ChatGPT subscriptions
- $C_{w,h}$: Wage Cost per hour
- $C_{w,m}$: Monthly gross wage cost

3.3.2 Overhead consumption and emission

To calculate the consumption quantities per person and per hour worked, the costs associated with a given overhead were divided by the unit price to obtain the consumption quantity [67].

(4) Overhead consumption per person per hour

$$O_{o,p/h} = \frac{C_{o,p/h}}{P_o}$$

Environmental calculations are limited to the inference phase of AI usage, focussing on the operational energy consumption and emissions during the application of ChatGPT in HR processes. Other life cycle stages, such as model training, hardware production, distribution, and end-of-life, are beyond the scope of this study. This is

one of the limitations of the study; however, the determination of the emission rate for the whole life cycle of the AI model is very difficult as it depends on many factors, and companies do not provide all the necessary information to be able to determine it. In the present case, the objective was to determine the emissions generated during the LCA use phase in Phase 4 for a period of 1 year. The first step was to determine the energy consumption of ChatGPT requests, which was calculated using a report published by Statista in 2024, which is 2.9 W-hours per request. In this case study, this value was used in the calculations and converted to kilowatt hours (0.0029 kWh) [69]. After determining the energy consumption of ChatGPT requests, the next step is to calculate how much $CO_{2\text{-eq}}$ the given amount of kWh corresponds to. To determine the emission rate, data from Electricity Maps [70] database was used, giving an overall emission of 230 g/kWh $CO_{2\text{-eq}}$ in February 2025 in California, USA [70]. Finally, by multiplying this by the number of requests generated in a year, the total annual CO_2-equivalent emissions from ChatGPT usage can be obtained. In this case study, the number of ChatGPT requests was determined based on the experience of the HR expert interviewed.

(5) $CO_{2\text{-eq}}$ of ChatGPT requests

$$\mathrm{CO}_{2GPT,y} = \left(\frac{E_{GPT,wh}}{1000}\right) \times EF_e \times N_{r,y}$$

In addition to the ChatGPT request output, the overhead consumption output is also defined for a year. For this purpose, it is necessary to multiply the amount of emission by the emission factor. To determine energy demand emissions, Electricity Maps (2025) were used, according to which the average emission in Hungary in February 2025 is 308 g/kWh $CO_{2\text{-eq}}$ [70]. To determine the emissions of district heat and gas in Hungary, the Hungarian Decree 9/2023 (25 May 2023) of the Hungarian Ministry of Economy and Labour [71] was used, which states that natural gas emits 297 g/kWh $CO_{2\text{-eq}}$and district heat 374 g/kWh $CO_{2\text{-eq}}$.

(6) $CO_{2\text{-eq}}$ of Overhead Consumption

$$\mathrm{CO}_{2O,y} = OC_y \times \mathrm{EF}$$

where:

- $C_{o,p/h}$: Overhead cost per person per hour
- $O_{o,p/h}$: Overhead consumption per person per hour
- P_o: Price per unit of overhead
- $CO_{2GPT,y}$ = Total annual CO_2-equivalent emissions from ChatGPT use (g $CO_{2\text{-eq}}$)
- $E_{GPT,wh}$ = Energy consumption per ChatGPT requests (Wh)
- $N_{r,y}$ = Number of ChatGPT requests per year
- EF_e = Emission factor of electricity (g $CO_{2\text{-eq}}$/kWh)
- $CO_{2O,y}$ = Total annual CO_2-equivalent emissions from overhead energy consumption (g $CO_{2\text{-eq}}$)
- OC_y = Total annual overhead energy consumption (kWh, m^3, or GJ depending on energy type)

- EF = Emission factor for overhead energy consumption (g $CO_{2\text{-eq}}$ per unit, eg, g $CO_{2\text{-eq}}$/kWh for electricity, g $CO_{2\text{-eq}}/m^3$ for gas, g $CO_{2\text{-eq}}$/GJ for district heating)

4 Results

The results section is divided into 2 main sections, where the qualitative (interview) and quantitative (case study) results are analysed.

4.1 In-depth interview

The in-depth interview provided essential qualitative input that formed the basis for both the process mapping and the subsequent quantitative analysis. The interviewee, a senior HR manager with experience in AI-assisted recruitment, outlined the recruitment process in detail, identifying 31 distinct steps involved from initial workforce planning to onboarding. These steps were validated and organised into a BPM, which can be seen in Fig. 4.

The recruitment process consists of six main phases that are built on each other to ensure the selection and integration of the right workforce into the organisation. The first step is the identification of recruitment needs, which includes annual HR planning and an organisational needs assessment, during which the needs received from managers are collected and processed. Then, the required job profiles are identified, budgets are planned, and a recruitment strategy is developed. The second stage is the formulation of the job description and expectations, during which the purpose of the job, its tasks, the skills, experience, and skills required are precisely defined. HR professionals then write the job advertisement, which is necessary to select the recruitment channels. The third step is the selection of the recruitment channels, which aims to determine the platforms on which the job advertisement for the vacancy should appear. At this stage, internal and external advertising is managed separately and the job vacancy is advertised through the appropriate channels. The fourth stage is the reception and pre-screening of applications, where the CVs and covering letters received are assessed. Following the pre-screening process, the most promising candidates are selected and contacted by email to arrange interview dates. The first round of interviews is conducted by the HR department using competency-based methods and, after evaluation, the candidates who pass to the second round will be interviewed on a professional basis. The fifth step is the selection and offer stage, during which the candidate is finalised on the basis of the results of the second-round interviews. Once a decision is made, the selected person is

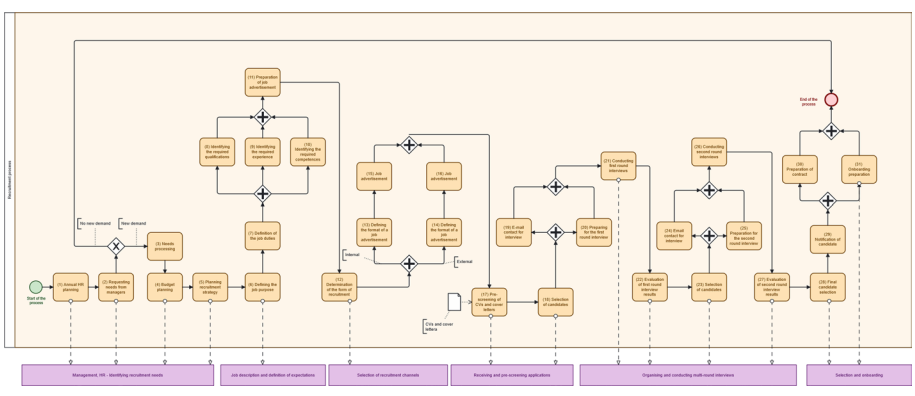

Fig. 4 Business Process Modelling of the Recruitment Process, own edit (Camunda Modeler)

notified, a contract of employment is drawn up, and preparations for the onboarding process are started to ensure a smooth integration into the new working environment.

During the interview, the participant not only the sequence of tasks, but also elaborated on how GAI has been integrated into various stages of the process. The interviewee reported that ChatGPT was most useful for writing job descriptions, screening candidate documents, and preparing interview questions and email communications. These stages were characterised by significant time savings and perceived gains in consistency and quality of communication.

Table 3 shows the time required for the recruitment process in hours and the number of ChatGPT messages involved as a case study for the organisation under study. Out of the 31 process steps, the interviewee found the use of GAI useful for a total of 14 activities, as it was possible to increase efficiency and thus reduce time in these activities. The table shows two scenarios, the first when no GAI is applied in the execution of activities, which for a recruitment process takes 73.5 h based on the case study. The second scenario, where GAI generated content is used in some process steps, takes 60.25 h. Thus, in the case study under consideration, a reduction of 13.25 h is achieved, requiring roughly 200 generated messages. The number of messages generated by GAI can certainly also be quite different, depending on the experience of the expert, as the quality of the prompts can greatly affect the output, and thus the number of iterations can also vary greatly. By formulating the right prompt, the GAI should be asked fewer questions, as it may generate the right answer the first time. ChatGPT and other GAI models can be useful, among other things, annual HR planning, helping to plan a recruitment strategy, draughting the text of a job advertisement, preparing for an interview, as for well as generating questions or preparing rehearsal tasks.

The qualitative analysis also revealed key themes regarding the perceived benefits and limitations of using generative AI in HR workflows. In particular, the respondent highlighted:

- Efficiency and time savings in routine administrative tasks;
- Increased standardisation and professionalism in communication;
- Concerns about overreliance on automated outputs and the importance of human oversight;
- Initial scepticism within the team, which gradually decreased after visible time-saving benefits emerged.

These insights enrich the quantitative findings by contextualising the observed differences in cost, time, and emissions between the two recruitment scenarios. They also provide a nuanced understanding of how GAI adoption is perceived at the practitioner level.

4.2 Case study

This section presents the findings of the single descriptive case study that examined how GAI integration of GAI impacted a specific organisational recruitment process. The objective of this case study was to examine how GAI integration of GAI influenced recruitment efficiency, energy consumption, CO_2-equivalent emissions, and related costs. To this end, two recruitment scenarios were compared: one using traditional methods without ChatGPT and another supported by GAI.

Table 3 Recruitment process

Recruitment process	Case 1—Time without ChatGPT (h)	Case 2—Time with ChatGPT (h)	Number of Responses by ChatGPT
(1) Annual HR planning	40	35	30
(2) Requesting needs from managers	0.5	0.25	5
(3) Needs processing	1	1	0
(4) Budget planning	1	1	0
(5) Planning recruitment strategy	2	1	20
(6) Defining the job purpose	0.5	0.25	5
(7) Definition of the job duties	1	0.5	10
(8) Identifying the required qualifications	1	0.5	10
(9) Identifying the required experience			
(10) Identifying the required competences			
(11) Preparation of job advertisement	1	0.5	10
(12) Determination of the form of recruitment	0.5	0.5	0
(13) Defining the format of a job advertisement (internal)	0.25	0.25	0
(14) Defining the format of a job advertisement (external)	0.5	0.5	0
(15) Job advertisement (internal)	0.25	0.25	0
(16) Job advertisement (external)	1	1	0
(17) Pre-screening of CVs and cover letters	4	2	50
(18) Selection of candidates	0.5	0.5	0
(19) E-mail contact for interview	1.5	1	10
(20) Preparation for the first round interview	1.5	1	10
(21) Conducting first round interviews	5	5	0
(22) Evaluation of the results of the first round interview results	1	1	0
(23) Selection of candidates	0.5	0.5	0
(24) Email contact for interview	1	0.5	5
(25) Preparation for the second round interview	2	1	20
(26) Conducting second round interviews	2.5	2.5	0
(27) Evaluation of second round interview results	1	1	0
(28) Final candidate selection	0.5	0.5	0
(29) Notification of candidate	0.5	0.25	5
(30) Preparation of contract	0.5	0.5	0
(31) Onboarding preparation	1	0.5	10
SZUM	73.5	60.25	200

The working hours for each case (without and with ChatGPT) reflect the sum of estimated time per task in 31 recruitment steps, based on the interviewee's experience. The values correspond to the total duration of a single recruitment process in each scenario

Sources: In-depth interview, own edit

4.2.1 Number of responses generated by ChatGPT number of working hours and the number of messages generated by the GAI per year

This subsection outlines the total number of working hours involved in the recruitment of 80 employees and the number of messages generated by ChatGPT in the GAI-supported scenario. Table 4 summarises the total time spent on recruitment activities with and without ChatGPT per year and quantifies the number of AI-generated responses based on process-level estimates.

The organisation under study had 80 new employees in 2024, so the number of hours and messages was multiplied by 80 to get the number of messages per year. ChatGPT use reduced the total recruitment time from 5,880 h to 4,820 h annually. Based on these figures and the estimated average number of ChatGPT messages per employee, the

Table 4 Recruitment process (1 year)

Number of new employees per year	Time without Chatgpt (h)	Time with ChatGPT (h)	Number of responses by ChatGPT (db)
80	5880	4820	16,000

Values reflect annual totals for 80 recruits. ChatGPT responses were estimated by multiplying average messages per recruit. These data serve as a basis for energy, emission, and cost analysis

Sources: Own calculations

Table 5 Overhead consumption (1 year)

	Gas (m^3)	Heating (Gj)	Electricity (kWh)	Water (m^3)
Without ChatGPT	2604.46	106.38	17,648.51	453.98
With ChatGPT	2134.95	87.20	14,466.98	372.14

Heating, gas, electricity, and water consumption values were calculated proportionally to the total working hours per scenario. The use of ChatGPT reduced working time by approximately 18%, resulting in corresponding reductions in resource usage

Sources: Own calculations

model generated a total of 16,000 messages over the course of a year. These data serve as the basis for further calculations of energy use, emissions, and cost.

4.2.2 Amount of overhead consumption in the organisation

Based on data from the organisation and annual accounts, we calculated the amount and cost of overhead associated with 1 h of work per employee. This allows one to determine the values based on the hourly demand of each work process.

Table 5 shows the amount of overhead, gas, heating, electricity, and water required by the recruitment process without using ChatGPT in the first case and with ChatGPT in the second case. The results shown in the table are calculated over 1 year based on the recruitment of 80 new colleagues. The table shows that since the second case is more efficient, the recruitment process can be carried out in less time, and therefore the amount of overheads is less.

4.2.3 Emission ($CO_{2\text{-}eq}$)

After the quantities related to the overheads were determined, the $CO_{2\text{-eq}}$ emissions were calculated. For the calculations, emissions could be assigned to three items, gas, heating, and electricity. The highest emissions are related to heating, with more than 11 million grammes of $CO_{2\text{-eq}}$, but if ChatGPT is also applied in the processes, about 2 million grammes of $CO_{2\text{-eq}}$ can be saved. For emissions related to electricity, about 1 million grammes of $CO_{2\text{-eq}}$ emissions can be saved and for gas, about 1.5 million grammes when the two scenarios are compared (Table 6).

For one year, the recruitment process requires roughly 16,000 messages generated by the model according to the case study, which requires 46.40 kWh of energy. This amount of kWh is equivalent to 10,672.00 grammes of $CO_{2\text{-eq.}}$ Later in the scenario in which LLM is applied in the process, we will adjust the result with this amount of emissions (Table 7).

4.2.4 Wage and overheads costs

In terms of costs, two types of costs have been calculated; one is personal income, which is the wage cost, and the other is the cost of the ChatGPT subscription, which has been

Table 6 Overhead emission

	$CO_{2\text{-eq}}$ (g)		
	Gas	**Heating**	**Electricity**
Without ChatGPT	8,261,256.92	11,051,860.95	5,435,742.27
With ChatGPT	6,771,982.71	9,059,518.67	4,455,829.55

$CO_{2\text{-eq}}$ emissions were calculated from overhead consumption data for gas, heating, and electricity. Values represent estimated annual emissions for 80 recruitments

Sources: Own calculations

Table 7 ChatGPT request emission

	Electricity (kWh)	**$CO_{2\text{-eq}}$ (g)**
ChatGPT request	46.40	10,672.00

Sources: Own calculations

Table 8 Average wage cost 2025

Hays Hungary Salary Guide	**Average gross wage**	**Gross average hourly wage**
Recruitment manager	€ 3500.00	€ 21.88
Recruitment partner	€ 3000.00	€ 18.75
IT Recruitment Specialist	€ 3000.00	€ 18.75
Recruitment Specialist	€ 2200.00	€ 13.75
Average	€ 2925.00	€ 18.28

Sources: Own calculations

Table 9 Wage cost (1 year)

Wage cost (without using ChatGPT, 1 year)	€ 107,493.75
Wage cost (with ChatGPT, 1 year)	€ 88,115.63
ChatGPT subscription (1 year)	€ 1448

Wage costs are based on time savings from using ChatGPT in recruitment. Subscription covers 5 users annually

Sources: Own calculations

adjusted for in the second scenario where ChatGPT is used in the process. The currency used for the calculations was EUR, which was 0.0025 EUR/HUF based on the exchange rate of 26.02.2025.

Table 8 shows the average salaries for recruitment jobs in 2025 as determined by Hays Hungary. In the calculation, we have included all positions that deal with recruitment at some level and have calculated the average hourly wages for the four selected positions below. According to the report, the average gross wage of a person working in recruitment in Hungary in 2025 is € 2925.00 and the average gross hourly wage is € 18.28 [68].

Table 9 shows the annual wage costs for the two cases and the ChatGPT subscription fee. The wage costs were calculated after determining the average hourly wage for 1 year. It is important to note that this cost is only the cost of the recruitment process and does not include the costs related to other HR tasks. In the case where the HR staff does not use ChatGPT, the gross wage cost would be € 1,07,493.75, but in the case where they use ChatGPT, it would be reduced to € 88,115.63, as the process can be completed sooner. In the case study we calculated 5 ChatGPT subscriptions, which can be used by several colleagues at the same time. Even if the wage cost is adjusted by the ChatGPT subscription fee, the cost is still more than € 17,000 in the scenario where ChatGPT is not used.

Table 10 Overhead cost

	Gas	Heating	Electricity	Water
Without ChatGPT	€ 1593.87	€ 4659.51	€ 2625.22	€ 762.35
With ChatGPT	€ 1306.54	€ 3819.53	€ 2151.96	€ 624.92

Overhead costs reflect proportional savings due to reduced HR working hours when using ChatGPT

Sources: Own calculations

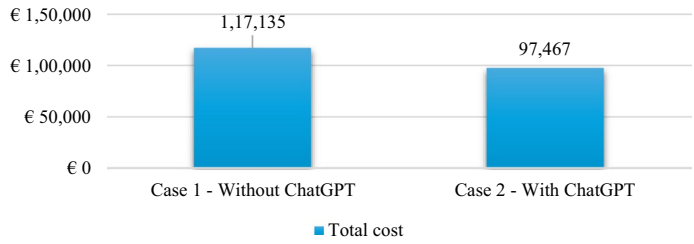

Fig. 5 Total cost, own edit

In addition to overhead volumes, reductions in overhead costs can also be achieved as a result of a reduction in working hours. The largest savings are also in heating, but reductions can also be achieved in the other overheads (Table 10).

5 Discussion

The purpose of the investigation was to examine the effectiveness and the environmental impact of the use of GAI in HR recruitment processes. The results show that GAI can result in significant time and cost savings while reducing the environmental impact. In the following, conclusions are drawn from the results, research questions are answered, and limitations of the research are described.

Figure 5 shows a summary of the costs related to the recruitment process, comparing the two cases. In the first case, the cost items include the wage cost and the overhead cost, and in the second case, the wage cost,, the overhead cost and the ChatGPT pre-recruitment cost over 1 year. Without using ChatGPT, the wage and overhead costs are € 117,135, while using ChatGPT they are € 97,467. This could result in savings of € 19,670. These savings will of course be more noticeable in the long term, when less human labour, for example, a smaller facility will be enough to complete the work.

Figure 6 shows the aggregated emissions for the two cases studied. In the first case, emissions are 24 700 thousand grammes of $CO_{2\text{-eq}}$, which can be reduced to more than 20 200 thousand grammes.

RQ1 sought to answer the question of how HR recruitment processes can use GAI to increase efficiency and reduce costs. The results of the research show that GAI can contribute to making the recruitment process more efficient and reducing costs in several areas. In particular, it has been shown to be useful in the preparation of job advertisements, the identification of required qualifications, the pre-screening of CVs, the preparation for interviews, and communication with candidates (e.g., by generating email templates). The case study showed that the use of GAI resulted in a reduction of 13.25 man-hours, which directly reduced salary and operational costs.

The second research question (RQ2) examined how the use of generative AI in recruitment processes affects the environmental performance of HR operations, specifically focussing on energy consumption and emissions. The case study examined in

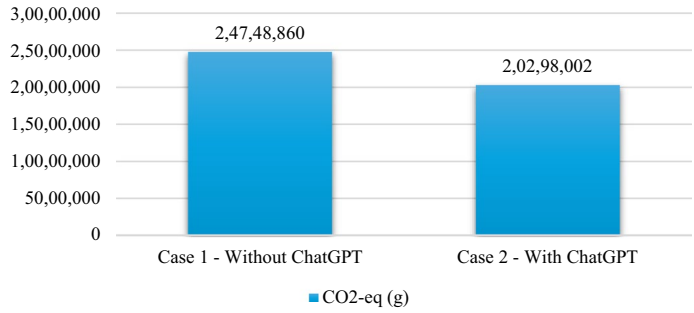

Fig. 6 Total emission, own edit

this research found that the use of GAI reduces energy consumption and emissions in recruitment processes, but is not entirely environmentally neutral. The energy consumption of the recruitment process reduced CO_2-equivalent emissions by 4.5 million grammes per year when using GAI. However, using the AI model still consumed 46.40 kWh of energy, resulting in 10,672 grammes of CO_2-equivalent emissions. This shows that while GAI can contribute to emission reduction in the short term, in the long term, more energy efficient AI models and sustainable data centre infrastructures are needed for AI to be a truly sustainable solution.

To further contextualise our findings, it is useful to compare them with previous empirical studies. The efficiency gains observed in our case study align with Abdelhay [18] SEM-based research, which also identified reduced bias, increased recruitment accuracy, and overall process acceleration when using generative AI tools. Similarly, the study by Hewage [19] concluded that AI systems such as ChatGPT are used more effectively in the early stages of recruitment, including CV pre-screening and administrative communication, areas where our case also reported the most significant time savings. Furthermore, Horodyski's [50] analysis using the UTAUT model supports our result that performance expectancy is a key factor influencing the adoption of AI tools in HR, consistent with our interviewee's positive perceptions of GAI. Although our research extends these findings by incorporating environmental dimensions (eg, energy use, $CO_{2\text{-eq}}$ emissions), it reinforces the growing empirical evidence that GAI can be a valuable tool in making recruitment more efficient and sustainable under the right conditions.

This study has several limitations that must be acknowledged. First, it focusses exclusively on recruitment and does not explore other HR functions such as training, engagement, or performance evaluation. The environmental analysis includes only the inference phase of AI use, omitting upstream impacts such as model training or hardware production. Furthermore, the findings are based on a single case study within a university-affiliated, limiting generalizability across sectors. Time estimates and efficiency gains are based on the subjective assessments of an HR expert, which may introduce bias. The study did not perform a formal sensitivity analysis to assess the impact of input variability on the calculated environmental and economic indicators. This represents a methodological limitation, as key parameters such as time estimates, overhead consumption values, and energy emission factors can be uncertain. As a result, the reported figures should be interpreted as indicative rather than absolute values. Furthermore, the application of ChatGPT was manual and not system-integrated, which may differ from the implementations of AI at the enterprise level. Human factors such

as adoption resistance, learning curves, and organisational readiness were not explored but could significantly influence outcomes in real-world deployments. Although the study briefly addresses labour costs in the context of efficiency, it does not assess wages or employment aspects as indicators of social sustainability. This is due to the study's primary focus on the environmental implications of GAI integration. Future research should explore how wage structures, job quality, and algorithmic fairness contribute to a more holistic sustainability assessment in HRM contexts.

Future research should further investigate how GAI affects employee acceptance, participation in the hiring process, and long-term HR strategies. In addition, a comprehensive understanding of AI's ecological footprint could be facilitated by a full life cycle analysis (LCA), particularly taking into account the environmental impacts of model training and hardware manufacturing. Further research could use longer-term longitudinal studies to map how the impact of GAI on organisational processes and workforce structure changes over time. It would also be useful to perform comparative analyses of multiple case studies in different industries and organisational types to improve generalizability. Furthermore, it would be important to explore the role of organisational culture, preparedness, and ethical perceptions, which can be key determinants of a company's success in integrating AI into its HR processes.

6 Conclusion

In the present study, environmental and effectiveness aspects of the use of GAI in HR recruitment processes were investigated. A case study was conducted comparing traditional and AI-based recruitment processes. The analysis not only looked at the time and cost efficiency of the processes, but also took into account the energy consumption and carbon emissions of GAI, giving a comprehensive picture of the environmental aspects of the technology. The results showed that the introduction of GAI into recruitment processes could lead to significant efficiency gains, reducing overall recruitment time by 13.25 h, which could mean savings of up to thousands of man hours per year. This could result in significant savings in labour and overhead costs for organisations. Cost analysis shows that AI-based recruitment methods can save an organisation more than €17,000 per year. Furthermore, by optimising processes, GAI contributes to sustainability goals (SDG 8, SDG 9, SDG 13) by reducing energy consumption and the environmental footprint of work processes. According to the analysis, the traditional recruitment process generates more than 24.7 million grammes of CO_2-equivalent emissions per year, while the GAI-based solution reduces this figure to 20.3 million grammes, a reduction of around 18%.

These results have important managerial and policy implications. From a managerial perspective, they offer HR professionals the opportunity to develop more efficient and sustainable recruitment strategies that reduce time and resource needs while improving overall process consistency. The use of generative AI can also reduce the subjectivity of recruitment decisions, potentially enhancing equal opportunity and fairness in candidate evaluation. At the same time, ethical and legal considerations must be addressed, particularly with respect to personal data protection and algorithmic transparency.

From a policy point of view, the growing integration of GAI into organisational workflows highlights the urgent need for clear regulatory guidance and standardised frameworks. Policymakers should support institutions in balancing innovation with

accountability by developing sector-specific AI governance models that include transparency obligations, human oversight requirements, and sustainability reporting. Furthermore, environmental regulations and digital transformation strategies could be expanded to recognise the carbon footprint of AI tools in administrative processes. Aligning the deployment of generative AI with data ethics, labor rights, and climate goals is essential to ensure long-term, responsible adoption at both the organizational and societal levels.

The application of generative AI in HRM raises ethical concerns related to transparency, fairness, and data protection. Algorithmic decision making can lack explainability and may reinforce biases if not regularly monitored. To mitigate these risks, organisations should implement governance mechanisms such as bias audits, human oversight protocols, and compliance with the GDPR and the EU AI Act. Ensuring transparency and fairness in AI-supported recruitment is essential to maintain trust and accountability.

To conclude, the research achieved its main objective to empirically assess the environmental and cost impacts of integrating GAI into HR recruitment processes in a real organisational setting. Through a descriptive case study approach, the study provided quantifiable insights into how GAI adoption affects time efficiency, carbon emissions, and financial costs. These findings contribute to bridging the existing gap in the literature by offering a multidisciplinary and data-driven perspective on the environmental impact of GAI in HRM. Overall, the study pointed out that GAI can increase the efficiency of HR recruitment processes while reducing energy consumption and environmental impact, thus playing an important role in promoting sustainable corporate operations.

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Data availability
The author confirms that all data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate
All methods were carried out in accordance with the Scientific Ethics Code of Széchenyi István University. No personal or sensitive data were collected or stored. Informed consent was obtained from the interviewee, and no formal ethical approval was required.

Consent for publication
Not applicable.

Competing interest
The authors declare no competing interests.

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